Do LLM credit-risk explanations stay faithful to SHAP?

300 high-risk loan applicants × 3 prompt conditions = 900 explanations. Click a condition to filter; pick any applicant below.

300	900	0	100%	35%
applicants	LLM explanations	invented features	top-3 found, applicant-level	top-3 found, no grounding

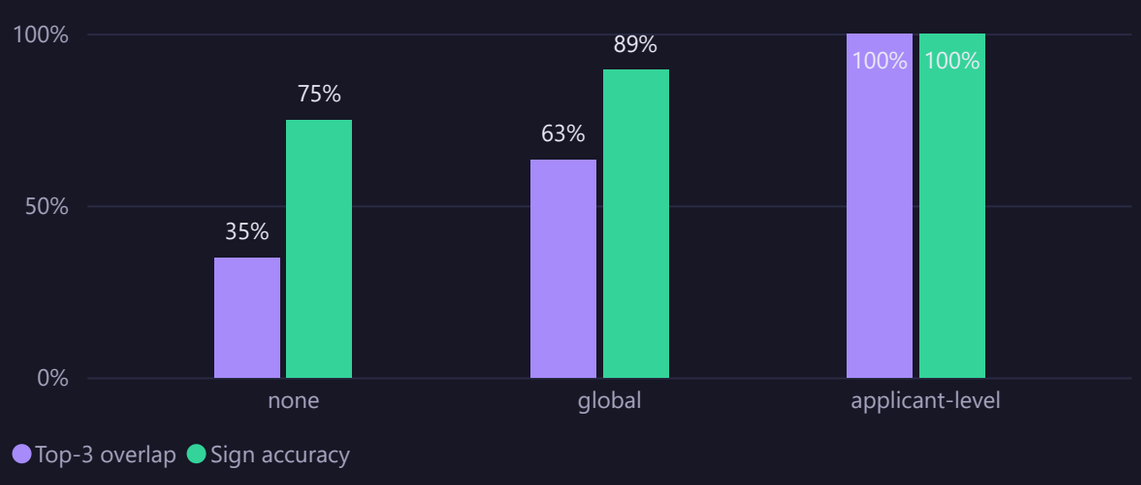
Takeaway: grounding the LLM in each applicant's own SHAP values recovers the most true top-3 drivers; with no grounding it finds the fewest (exact figures in the cards above).

Prompt condition (click to filter the charts below)

applicant-level | global | none

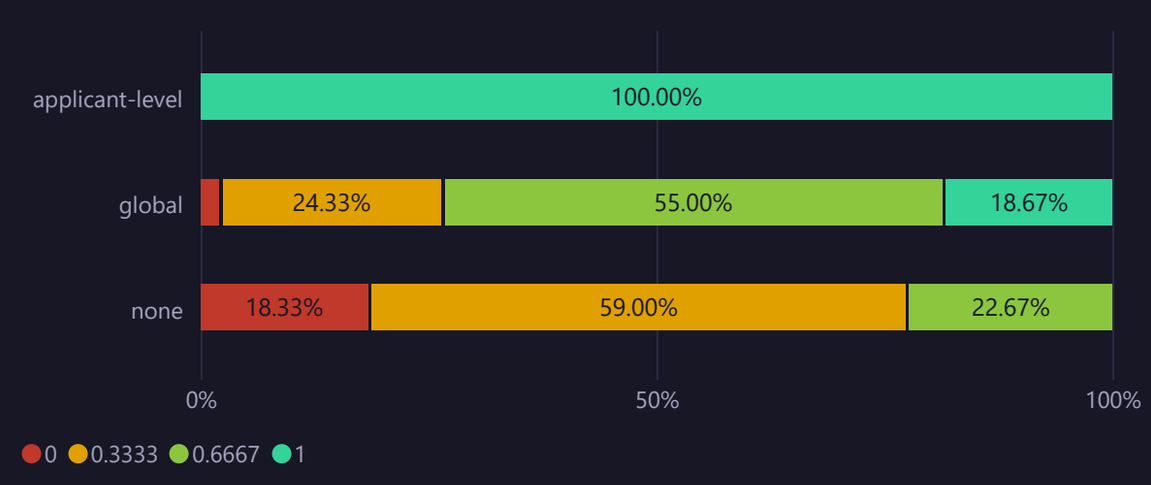
Average accuracy by condition

Top-3 overlap and Sign accuracy by condition



How many of the true top-3 drivers did the LLM name?

Applicants by condition and Share of true top-3 named



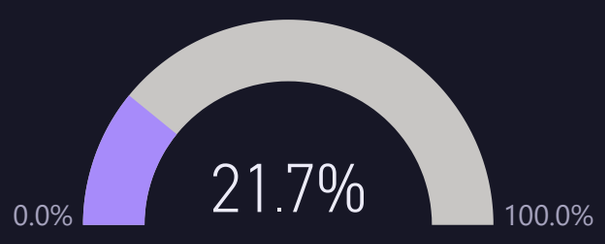
Explore one applicant

Applicant (300 high-risk)

269846

Predicted default risk (selected applicant)

predicted default risk (pick an applicant)

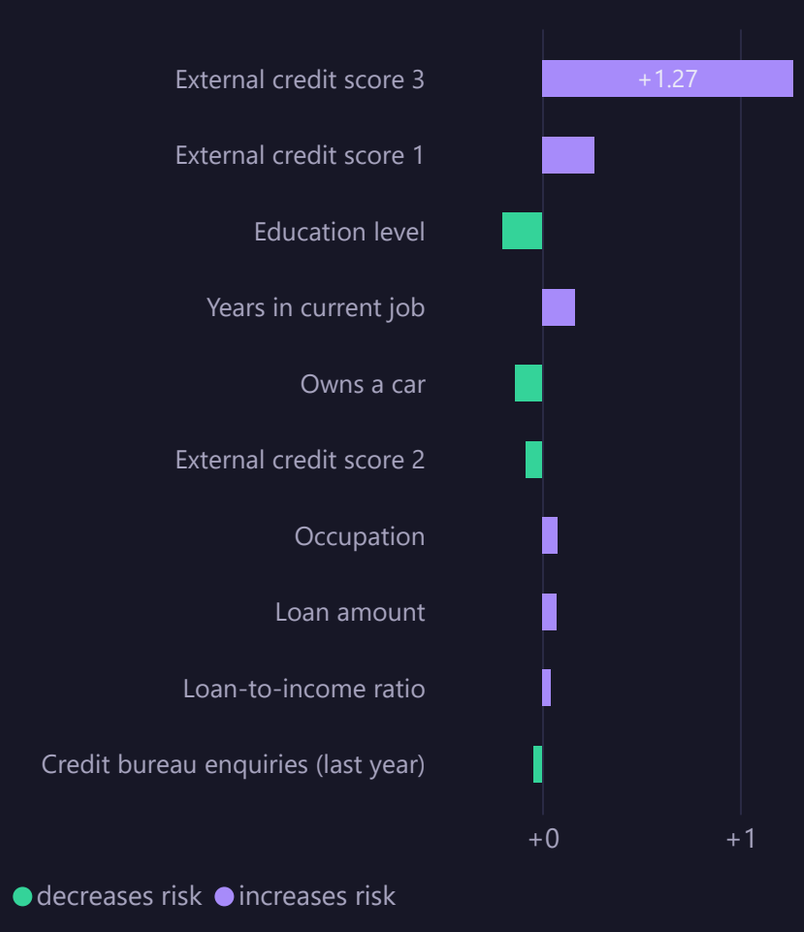


Applicants where "global" missed a true driver — copy an ID into the slicer

Applicant	Top-3 overlap	Sign accuracy
115629	0%	100%
183469	0%	100%
201475	0%	67%
337470	0%	100%
341262	0%	67%
408004	0%	100%
107905	33%	67%

What the model really used (SHAP, top 10)

SHAP value, SHAP rank (1 = strongest) and Feature value by...



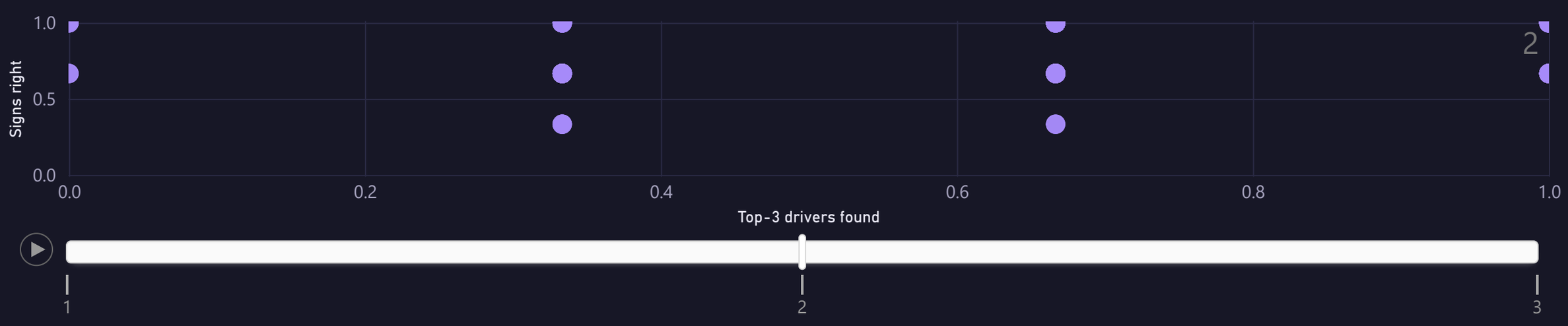
What the LLM said (for the selected applicant)

#	Condition	Top-3 found	Signs right	Invented	What the LLM said
1	none	33%	100%	0	The applicant's loan amount is almost three times their annual income, which is a strong signal of higher default risk. Additionally, they have been in their current job for less than a year, indicating limited employment stability. Their external credit score is also low, further raising the likelihood of repayment problems.
2	global	67%	67%	0	The applicant's three most influential risk factors are the external credit scores, all of which are relatively low (0.1, 0.56, and 0.26). These low scores signal a higher likelihood of default, outweighing the otherwise solid income and asset profile.
3	applicant-level	100%	100%	0	The model flags this applicant as high risk mainly because two of the external credit scores (score 3 and score 1) contribute strongly to a higher default probability. While the applicant's higher education slightly offsets the risk, it is not enough to outweigh the negative impact of the credit scores.

Watch grounding move every applicant: press play

Each dot = one applicant. Play: 1 = no grounding, 2 = global importance, 3 = applicant-level SHAP. Dots move right (more true drivers found) as grounding improves.

Top-3 drivers found and Signs right by applicant_id and condition_order



Hypothesis tests (significant = p < α; the three H3 tests use a Bonferroni α of 0.0167)

Hypothesis	Test	Effect-size type	Effect size	p-value	α	Significant?
H1 External credit score 2 vs default	Welch t-test	Cohen's d	-0.597	0.0000	0.0500	True
H1b Loan-to-income ratio vs default	Mann-Whitney U	rank-biserial r	-0.004	0.3188	0.0500	False
H2 Education level vs default	Chi-square	Cramer's V	0.058	0.0000	0.0500	True
H3 global vs applicant-level (top-3 overlap)	Wilcoxon signed-rank	mean gain in overlap	0.366	0.0000	0.0167	True
H3 none vs applicant-level (top-3 overlap)	Wilcoxon signed-rank	mean gain in	0.652	0.0000	0.0167	True